Course: Digital Logic Design (CSE 2323)
Semester: 3rd (SP_25)   Section: 3 AM,3CM
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Final Exam Preparation Guideline

Dear Students,
This guideline highlights the key topics and areas you need to focus on for your final exam in **Digital Logic Design (CSE 2323)**—review each section carefully and practice actively.

All lecture slides have been provided in the classroom and should be sufficient for your preparation. For deeper understanding, you may also refer to the attached textbook: *Digital Logic & Computer Design* by Morris Mano.

Combinational Circuits: Focus on look-ahead carry generators, 2-bit and 3-bit magnitude comparators, binary parallel adders, and encoders. Understand encoder limitations and how to overcome them. Be able to implement logic functions using decoders and construct higher-order decoders (4-to-16, 5-to-32) from lower ones (Ref: Morris Mano 5.2, 5.4, 5.5).

Multiplexers & Demultiplexers: Know different types of multiplexers, their structure, and how to implement Boolean functions, XOR gates, and full adders using MUX. Understand how to derive functions from multiplexers and the concept of demultiplexing.

Sequential Circuits: Study latch and flip-flop types (SR, JK, D, T), their characteristic tables, equations, and timing diagrams. Practice flip-flop conversions (SR to JK, JK to SR, SR to D,JK to D), and understand propagation delay, race-around condition, and master-slave flip-flops.

State Equations: Review examples 1 and 3 from slides. Be able to derive state equations, next-state tables, and transition diagrams based on given sequential circuit logic.

Shift Registers: Understand the operation and differences among SISO, SIPO, PISO, and PIPO shift registers. Know their practical applications and data movement process.

Counters: Practice designing 3-bit and decade synchronous counters. Learn how to implement arbitrary sequence counters using JK, T, or D flip-flops with appropriate state diagram and state table

Memory and Function Implementation: Understand ROM structure, its types, and how to implement logic functions using ROM. Study internal RAM decoding (Ref: Morris Mano 7.8) and function implementation using PLA, PAL

General Advice: Functions may be given in forms such as truth tables, minterms, decimals, or Boolean expressions—be prepared to implement them using ROM, PLA, PAL, MUX, or decoders. Practice circuit construction, diagrams, and implementation logic for each form.